

Houze Yang

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Google Scholar | LinkedIn

Research Focus: Computer vision and multimodal learning for gaze, social attention, and human behavior; interactive AI and extended reality.

Education

University of Illinois Urbana-Champaign

- **Ph.D. in Industrial Engineering** Aug 2025 – May 2030 (expected)
Advisor: Inki Kim GPA: 3.92/4.00
- **B.S./M.C.S. in Computer Science (4+1 Program)** Aug 2019 – May 2024
GPA: 3.97/4.00

Research Experience

Research Assistant

Aug 2025 – Present

Health Care Engineering Systems Center, University of Illinois Urbana-Champaign

GazeAnywhere

CVPR 2026, Equal Contribution

- Developed a data engine to extend existing gaze datasets with text prompts identifying whose gaze to estimate, integrating cross-dataset annotation alignment, VLM-based pseudo-annotation, and automated verification with human review.
- Built Gaze-Co from GazeFollow, VideoAttentionTarget, and ChildPlay, producing approximately 120K prompt-annotated training samples and concept-based test sets for evaluating promptable gaze target estimation.
- Fine-tuned open-vocabulary detection to supply text-conditioned head bounding boxes to conventional gaze-estimation baselines; evaluated the adapted pipelines and contributed training runs for selected ablations.

GazeAgent: Query-Driven Gaze Analysis

Project Lead

CVPR 2026 Demo (accepted); full manuscript under review at ACM IMWUT.

- Conceived and led GazeAgent to bridge the gap between spatial gaze predictions and behavioral analysis, enabling natural-language questions about attention targets, durations, and event sequences in naturalistic interactions.
- Designed a semantic gaze-event representation by grounding frame-level predictions in tracked scene entities and aggregating them over time, capturing object attention, eye contact, and gaze shifts.
- Built a code-generating agent that executes natural-language queries as Python analyses over the event timeline, separating perception from analysis and retaining event records, programs, and results for inspection and re-execution.
- Developed a 125-query benchmark and conducted a controlled user study ($N = 16$); achieved 83.3% temporal-context accuracy versus 66.7% for the strongest video-language baseline on this task, and reduced perceived workload during video review (NASA-TLX, $p = .001$).

Research Assistant

Jan 2024 – May 2025

Health Care Engineering Systems Center, University of Illinois Urbana-Champaign

- Developed a Magic Leap 2 application in Unity/OpenXR to record gaze trajectories and support visual-attention analysis during brain CT review.
- Built shader-based simulations of central and peripheral visual-field loss and blurred vision for AR user studies.

Research Assistant

Oct 2023 – Jan 2024

Smart Lab, University of Illinois Urbana-Champaign

- Explored indoor SLAM and 3D reconstruction with Cartographer, RGB-D/LiDAR sensing, and Instant-NGP; built workflows for occupancy mapping and real-time trajectory estimation.

Publications

Gaze Target Estimation Anywhere with Concepts

X. Cao*, **H. Yang***, V. Gunda, Z. Zhou, T. Xu, A. Kowdle, I. Kim, J. M. Rehg
CVPR 2026, pp. 31304–31315. *Equal contribution. [Code](#)

ChronoVision: Temporal Reasoning via Latent State Reconstruction

Y. Shen, J. Xu, B. Li, Y. Zhang, T. Yu, B. Li, **H. Yang**, R. Wang, X. Cao
EMNLP 2026 (accepted).

Integrating Machine Learning into Extended Reality: A Critical Review of Computational Approaches to Human Behavior Modeling

H. Yang, C. Cao, A. Hosny, J. Ryu, I. Kim
HCI International 2025, Late Breaking Papers, LNCS, pp. 49–59.

The 1st AI Children Challenge

B. Li, Y. Shen, **H. Yang**, X. Cao, G. Yun, L. Gao, T. Chen, L. Xu, J. Cao, M. Huang
CVPR Workshops 2026, pp. 5564–5570.

A Survey on Embodied AI for Pediatrics

Y. Shen, Q. Zhang, Y. Zhang, J. Zhou, **H. Yang**, Y. Wang, J. Zhu, O. K. Susladkar, Y. Tian, M. Huang, J. Chen, X. Cao
CVPR Workshops 2026.

Preprints

GRASP: Learning to Ground Social Reasoning in Multi-Person Non-Verbal Interactions

J. Kim, X. Cao, **H. Yang**, B. Boote, A. Jojic, F. Ryan, B. Lai, S. Lee, J. M. Rehg
arXiv:2605.15764, 2026.

EgoForge: Goal-Directed Egocentric World Simulator

Y. Shen, J. Liu, X. Li, Y. Liu, B. Li, **H. Yang**, W. Jia, Y. Li, T. Yu, J. M. Rehg, X. Cao, I. Lourentzou
arXiv:2603.20169, 2026.

Technical Skills

Programming & ML: Python, C#, PyTorch, vLLM, Linux

Research Methods: Vision-language modeling, video understanding, gaze estimation, semantic grounding, benchmark design, controlled user studies

XR Development: Unity, OpenXR, Android, Magic Leap 2, RayNeo X3 Pro

Additional Experience

Unity VR Game Developer

Oct 2021 – Sep 2022

Resilience Inc.

- Developed VR game features in Unity/C# and optimized runtime performance through scene, model, and code improvements.